Exercise #1: Topics in Theoretical Statistics

1- Let X be an absolute continuous random variable with the following cdf:

$$F(x) = 1 - e^{-x^2}, \quad x \geq 0.$$

A) By using an appropriate transformation, construct a new distribution that adds a scale parameter, say $\beta > 0$, to this distribution.

B) Generate 1000 data from the obtained distribution in A, with $\beta = 1.5$, and estimate the parameter β.

C) If the random variable X follows the distribution A, obtain the distribution of $Y = X^{\frac{2}{\alpha}}\beta^{\frac{2-\alpha}{\alpha}}$; $\alpha > 0$. What is the name of this distribution?

D) Estimate the parameters α and β in C by simulation for $n = 100, 200, 500,$ and 1000. Take the true values as $\alpha = 1.5, \beta = 0.5$ and $\alpha = 0.5, \beta = 1$. Provide a table for the results.

E) Plot the density function of the obtained distribution in **C**, for $\alpha = 2, \beta = 0.5$; $\alpha = 1, \beta = 1$; $\alpha = 0.5, \beta = 0.5$; and $\alpha = 0.5, \beta = 1.5$. All densities should be plotted on one graph.

2- Consider the absolutely random variable X with the following cdf:

$$F(x) = 1 - (1 - x^a)^b; \quad 0 < x < 1, \ (a, b > 0).$$

A) Plot the density function corresponding to the above cdf, for $a = 1, b = 1$; $a = 1, b = 2$; $a = 2, b = 1$; $a = 0.5, b = 2$; $a = 2, b = 0.5$; and $a = 0.5, b = 0.5$. (All in one graph).

B) Estimate the parameters a and b by simulation for $n = 1000$ and 5000. Take the true values $a = 2, b = 1$ and $a = 1.5, b = 1.5$. Replicate this procedure 10,000 times, for each sample size, and consider the mean of the estimates for the final estimation of the parameters.

C) Estimate the parameters using the attached real data set.

3- Consider the FGM copula $C(u,v) = uv(1+\alpha(1-u)(1-v)),\ |\alpha| \leq 1$. Construct a bivariate distribution with the marginal cdfs given in Exercise #2. Plot the joint density function of the obtained distribution. ■